

# Akshat Sharma

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## EDUCATION

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**Maharaja Agrasen Institute of Technology**

B.Tech - Electronics and Communication Engineering | **8.24 CGPA**

**Rohini, Delhi**

2023-2027

## EXPERIENCE

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### Embedded Systems Engineering Intern @ Rebhu Computing

- Developed FPGA based Matrix Multiplication Accelerators using Artix A7-100T for optimizing performance of AI models.
- Implemented **Bluetooth LE PAwR (Periodic Advertising with Responses)** protocol for scalable bidirectional communication between ESL (Electronic Shelf Label) tags and access points.
- Designed and developed **low-level device drivers** for SPI-based display modules, enabling efficient display updates and reduced power consumption.
- Worked on board bring-up, hardware validation, and peripheral integration for embedded platforms.

### Robotics Intern @ Rebhu Computing

- Developed an **autonomous UGV platform** with **ArduRover & Jetson Orin**.
- Created custom **ICP-based module for odometry estimation** from point cloud.
- Fused wheel encoder data (via **CAN bus**) and IMU data for enhanced localization.
- Worked on Ardupilot-Gazebo **SITL simulations**.
- Set up **RTSP** camera video capture pipelines using **GStreamer**.

### Open Source Hardware Intern @ FOSSEE, IIT Bombay

- Contributed to the integration of **ESP32 emulation** into FOSSEE's Yaksh automated evaluation platform.
- Worked with **QEMU**-based emulation environments for ESP32 firmware execution and validation.
- Developed and tested workflows supporting both **MicroPython** and **ESP-IDF** applications.
- Assisted in creating reproducible embedded software test environments for large-scale student evaluation.

## PROJECTS

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### Eyantra Robotics Competition '25 - Verilog

[Project Link](#)

- Made a **maze solving robot** using the **de0 Nano FPGA board**.
- Implemented **PWM generation, distance measurement** using **ultrasonic sensors, UART receiver and transmitter** and **DHT11 Humidity and Temperature** sensor interface.
- Designed and implemented a **RISC-V CPU** in Verilog supporting all **40 RV32I base instructions**.
- **Simulated, synthesized, and verified** the design using **Intel Quartus Prime and Modelsim**.

### WS2812B DMA-based Driver - STM32

[Project Link](#)

- Implemented a high-speed LED driver for **WS2812B** addressable LEDs on an **STM32F103C6T6** by configuring **hardware timers** for precise **PWM** signal generation.
- Optimized CPU utilization by offloading data transmission to the **DMA (Direct Memory Access)** controller, ensuring accurate timing for the single-wire serial protocol.
- Verified signals and timing on an **oscilloscope**.

### 3D Mesh Visualization & Embedded Display Streaming - ESP-IDF | SDL2

[Project Link](#)

- Implemented a **basic 3D renderer in C using SDL2** to project 3D (X, Y, Z) points onto a 2D screen.
- Wrote a simple **OBJ Parser** to retrieve faces and vertices from the file.
- Used **UDP sockets** to stream rendered frame data from a PC to an **ESP32**.
- Displayed the received stream on an **OLED screen** connected to the ESP32 via **SPI**.

## SKILLS

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**Languages:** C/C++ | Python | Verilog

**Technologies:** STM32 | ESP32 | BLE | Embedded Linux | ROS2 | Docker | Git | QEMU | KiCad | Fusion360

**Skills:** Embedded Systems | Bare-Metal Firmware | Device Drivers | FPGA Design | RISC-V | PCB Design | Robotics